

DTE, in Plain Terms

A website that puts household energy costs next to DTE Energy’s finances. Every number has a source. If a number is missing, the page says so.

This guide is the short version. It explains what the site does, how to run it, and where to look in the code.

1. What the site is

The site is independent. DTE Energy did not publish it.

It has two kinds of numbers:

- **Checked numbers** come from public agencies and from DTE’s own filings. They are saved in the project and shown on the main pages.
- **Uploads** are spreadsheets anyone can add. The site describes them and compares them with the checked numbers. Uploads are marked “not checked.”

2. The pages

PAGE	ADDRESS	WHAT YOU SEE
Home	/	Four big numbers: energy burden, the price of electricity, shutoffs, and CEO pay.
Household	/household	A neighborhood map, burden by income, shutoffs, and electricity prices.
Financials	/financials	Stock price, dividends, revenue growth, and CEO pay.
Compare	/compare	One household number and one company number on the same years.
Your data	/data	Upload a file, and see recent uploads.
Saved upload	/data/...	Charts and short findings for that file.

Energy burden means the share of a household’s income that goes to home energy. Above 6% is treated as high.

Two lines moving together does not prove that one caused the other. The Compare page says this in plain words.

3. Where the checked numbers come from

The site does not call those agencies every time someone opens a page. A script downloads the files once and saves one snapshot: `src/server/data/verified.json`.

TOPIC	SOURCE
Stock price, dividends, revenue growth	SEC filings and Yahoo Finance
CEO pay	DTE proxy statements, typed into a JSON file
Customers, sales, prices, bills	U.S. Energy Information Administration (Form EIA-861)
Shutoffs	Michigan Public Service Commission case U-18120, typed into a JSON file
Energy burden and the map	U.S. Department of Energy, LEAD tool, 2022
Michigan energy hardship	U.S. Census Household Pulse Survey

CEO pay and shutoff counts were copied from PDF reports by hand. Change those files only after checking the original document.

To refresh the snapshot:

```
npm run seed
```

To refresh one part only, add `--only=financial`, `eia`, `shutoffs`, `lead`, or `pulse`.

4. How to run it

You need Node.js. Then, in the project folder:

```
npm i
npm run dev
```

Open **`http://localhost:8080`**.

The snapshot already in the project is enough. You only run `npm run seed` when you want newer numbers. The first seed needs the internet. Later runs reuse downloaded files.

COMMAND	WHAT IT DOES
<code>npm run dev</code>	Start the site
<code>npm run seed</code>	Rebuild the snapshot of checked numbers
<code>npm test</code>	Test the upload analysis

COMMAND	WHAT IT DOES
<code>npm run typecheck</code>	Check the code for type errors
<code>npm run lint</code>	Check style
<code>npm run build</code>	Make the production build

5. What it is built with

Everything is TypeScript. There is no Python, no separate server app, and no database.

JOB	TOOL
Pages	React
Routes and server	TanStack Start. A file in <code>src/routes</code> is a page.
Loading data	TanStack Query. The page asks for data before it draws.
Dev server and build	Vite. The site runs on port 8080.
Production bundle	Nitro, aimed at Cloudflare. Local development stays on your computer.
Look and layout	Tailwind CSS. Colors are named tokens in <code>src/styles.css</code> .
Line and bar charts	Recharts
Neighborhood map	d3-geo, using a map file already in the project
Income bars	Plain HTML. No chart library.
Checking uploads	Zod
Reading spreadsheets	Papa Parse for CSV, SheetJS for Excel
Optional written summary	Vercel AI SDK. Skipped if there is no API key.
Tests	Vitest

The folder `src/components/ui` is a kit of buttons, menus, and dialogs. The real pages barely use it. The charts use only its chart wrapper.

6. How a page gets a number

1. The page file, for example `src/routes/household.tsx`, asks for its data before it renders.
2. That request goes to a function in `src/data/measures.ts`.
3. The function reads the snapshot through `src/server/measures.ts`.
4. The page draws the result with a chart and a source line.

Code under `src/server` stays on the server. The browser must not import it. The build fails if it does.

Each stored number is a short list:

```
[measure, period, grain, dimension, geoId, value]
```

`period` is a date like `2024-01-01`. `grain` is month, quarter, or year. `dimension` is a split such as "homes" or "electric." `geoId` is a neighborhood id, or empty when the number is for the whole area.

A year is shown only when every month in that year was reported. A half year is never drawn as if it were a full year.

7. Where to look in the code

Start with these files, in this order:

- 1. `README.md` — the longer project description
- 2. `CODEBASE.md` — more detail for people changing the code
- 3. `src/data/types.ts` — what a measure and a number look like
- 4. `src/data/measures.ts` — how a page’s data is prepared
- 5. `src/routes/household.tsx` — a typical page

FOLDER	WHAT IS IN IT
<code>src/routes</code>	The pages
<code>src/components/charts</code>	Charts and the map
<code>src/data</code>	Types, labels, and the functions pages call
<code>src/server</code>	Reading the snapshot, saving uploads, optional AI summary
<code>src/lib/ingest</code>	Reading an uploaded file and writing the findings
<code>scripts/seed</code>	Scripts that download or read each source
<code>public/geo</code>	The neighborhood shapes for the map

8. Uploads

On **Your data**, the browser reads the file first. The server checks the limits again. It does not trust the browser’s row count.

Accepted files: CSV, TSV, Excel (.xlsx), JSON, and one JSON object per line.

- Up to 10 MB, 20,000 rows, and 60 columns
- Up to 10 uploads an hour from the same address

The server then describes the columns, writes a trend, a ranking, and a spread, and lines years or places up with the checked DTE numbers. If an API key is set, it can add a short summary. Any sentence that invents a number is thrown away.

Files are saved on the computer running the site, in a folder named `.data/uploads`. They are not saved in git. Restarting the site clears the hourly limit. A hosted Cloudflare copy of the site has no disk for these files, so uploads work on your machine, not on that host, until a real store is added.

Do not upload names, addresses, phone numbers, or account numbers. Uploads can be opened by anyone who has the link.

The optional summary uses a file named `.env`. Copy `.env.example` and set `AI_GATEWAY_API_KEY`. Leave it empty and the rest of the site still works. `AI_MODEL` is optional. The default model is `openai/gpt-5.4-mini`.

9. Before you change something

- Do not fill in a missing year. Leave the gap.
- Do not treat an upload as an official figure.
- Format numbers with `src/lib/format.ts` and show the source with `src/lib/citation.ts`.
- Chart colors come from the `--chart-` names in `src/styles.css`. Do not hard-code colors in a component.
- In the line chart, the axes are passed as a list. Do not wrap them in a React fragment, or the axes vanish.
- Run `npm test`, `npm run typecheck`, and `npm run build` before you share a change. After a visual change, open the page in the browser.
- Do not rewrite history that is already on GitHub. No force push.

DTE, in Plain Terms — an independent project. Repository: github.com/tseten14/Energy-Equity-Project. Longer notes live in README.md and CODEBASE.md.